Full-Grid Subgroup Simulation Study under Marginal Non-Proportional Hazards

NPH Project

February 28, 2026

Abstract

We report a complete subgroup-scenario simulation study covering all protocol-grid combinations (1296 scenario cells, 200 replications per cell). Non-proportional hazards are induced at the marginal arm level by subgroup-mixture effects. We provide full-grid visualizations, one-factor-at-a-time (OFAT) slices for interpretability, and scenario-level recovery artifacts enabling exact lookup of each scenario result.

1 Introduction

Comparative performance of survival methods under non-proportional hazards (NPH) depends on the estimand and data-generating assumptions [5]. In subgroup contexts, marginal NPH can arise even when subgroup-level hazards are proportional. This paper presents full-grid evidence and an interpretation strategy that balances completeness and scenario-level recoverability.

2 Methods

2.1 Data-generating process

For control, $S_0(t) = \exp(-\lambda_0 t)$. For treatment,

$$S_1(t) = p \exp(-\lambda_0 HR_+ t) + (1-p) \exp(-\lambda_0 HR_- t). \tag{1}$$

Thus hazards are proportional within subgroup but non-proportional marginally after mixing over biomarker status.

2.2 Protocol grid and simulation size

The grid varies baseline hazard (3), censoring level (3), prevalence (3), overall effect-size level (4), subgroup-relative effect-size level (3), and sample size (4: 300/500/1000/1500), for $3 \times 3 \times 3 \times 4 \times 3 \times 4 = 1296$ cells. Each cell uses 200 replications.

2.3 Estimands and methods

RMST estimand at horizon τ:

$$\Delta_R(\tau) = \int_0^\tau \{S_1(u) - S_0(u)\}\, du, \qquad \tau \in \{6, 12\} \text{ months.} \tag{2}$$

Methods include log-rank, FH-weighted log-rank family, MaxCombo (Bonferroni minimum-p approximation), Cox PH, and RMST tests. AHR is estimated over observed support $t_{\max} = \max_i T_i$ when applicable.

2.4 Scenario recoverability

Every scenario is assigned a deterministic `scenario_id` in `results/scenario_dictionary.csv`. Full outputs are provided in wide and long formats: `scenario_metrics_wide.csv` and `scenario_metrics_long.cs` OFAT slices are provided in `results/ofat_*.csv`.

3 Results

3.1 Complete full-grid view (all results shown)

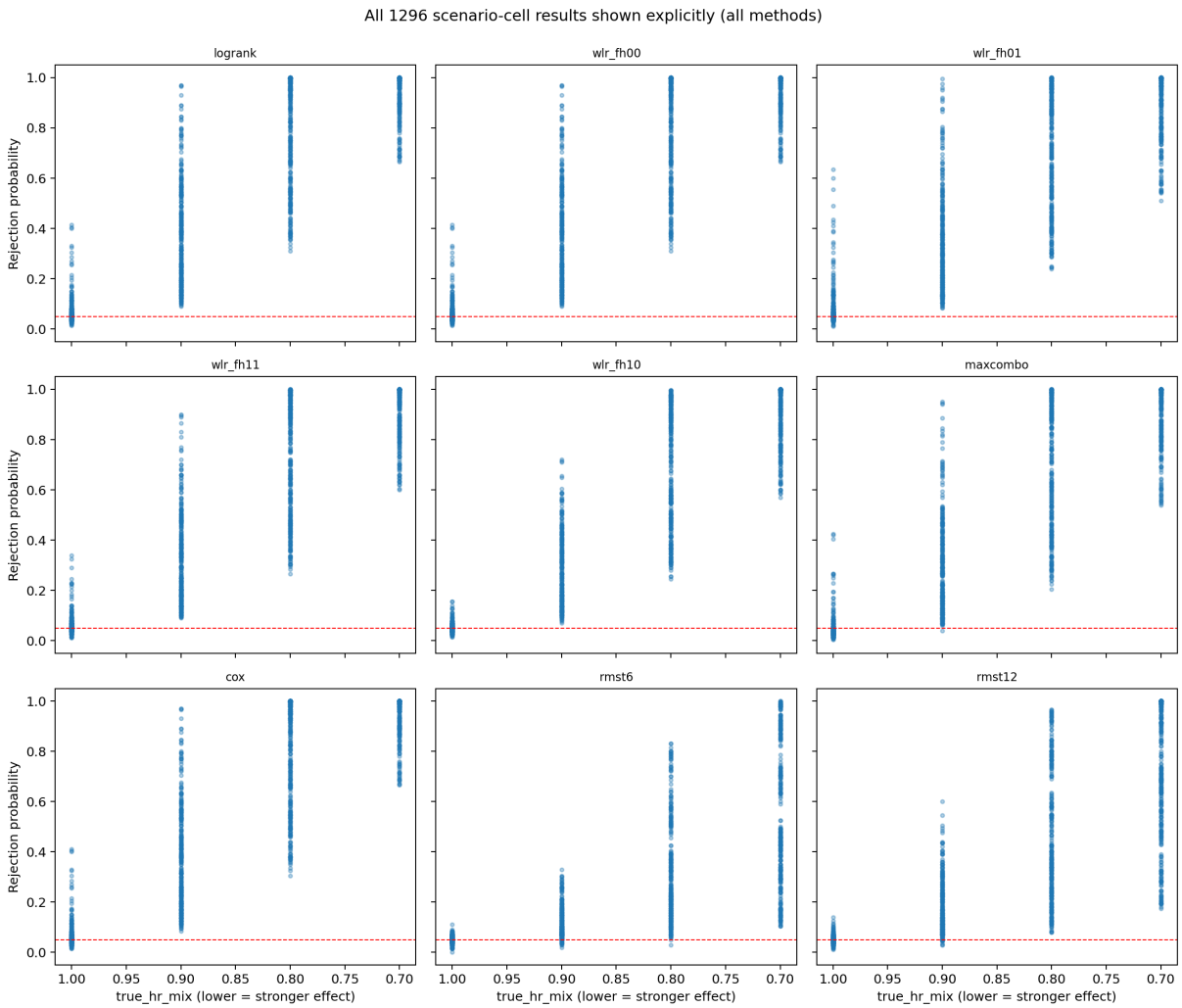

Figure 1: All 1296 scenario-cell results plotted for all testing methods (no scenario omission).

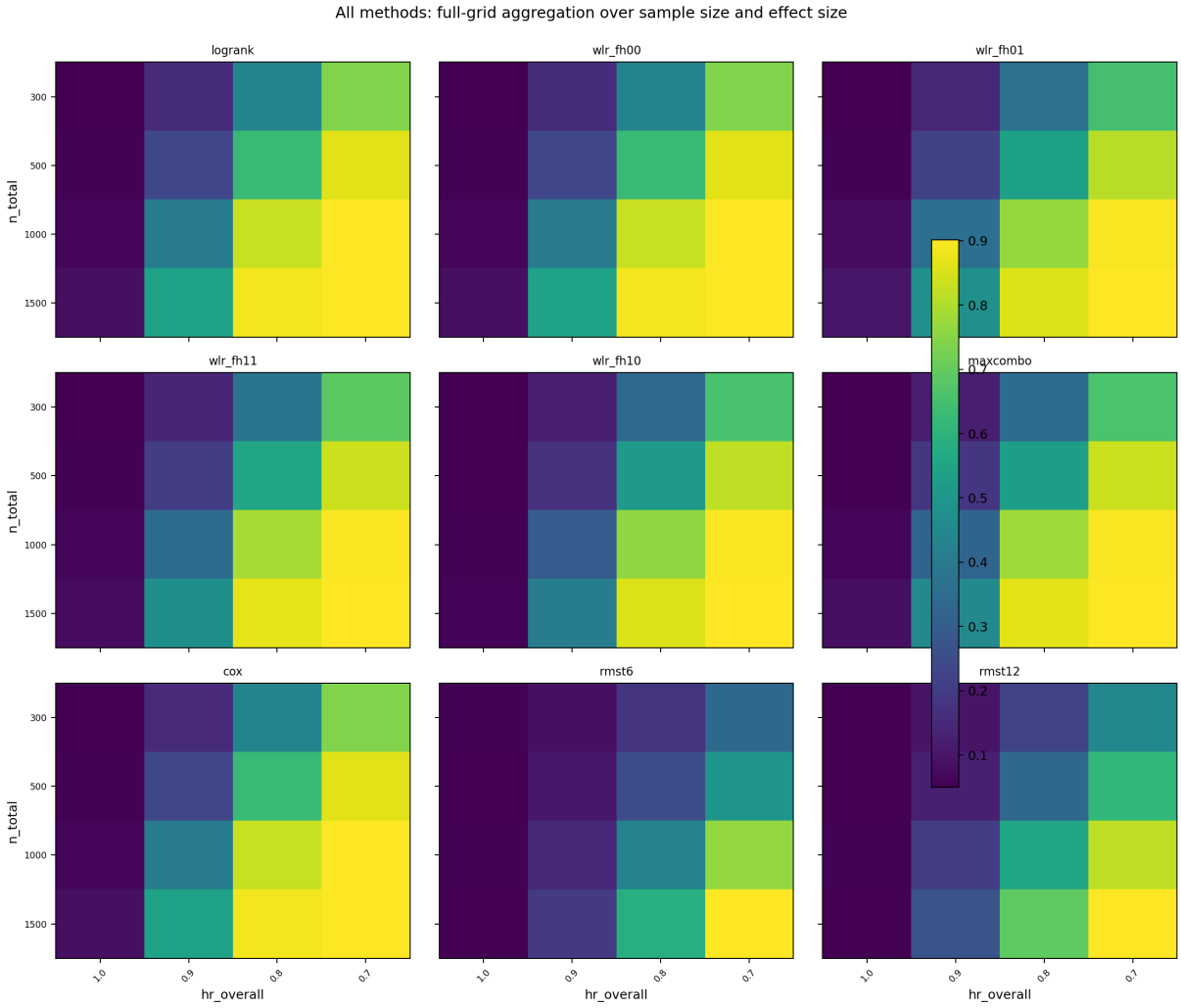

Figure 2: Full-grid method heatmaps aggregated over sample size and overall effect-size axes.

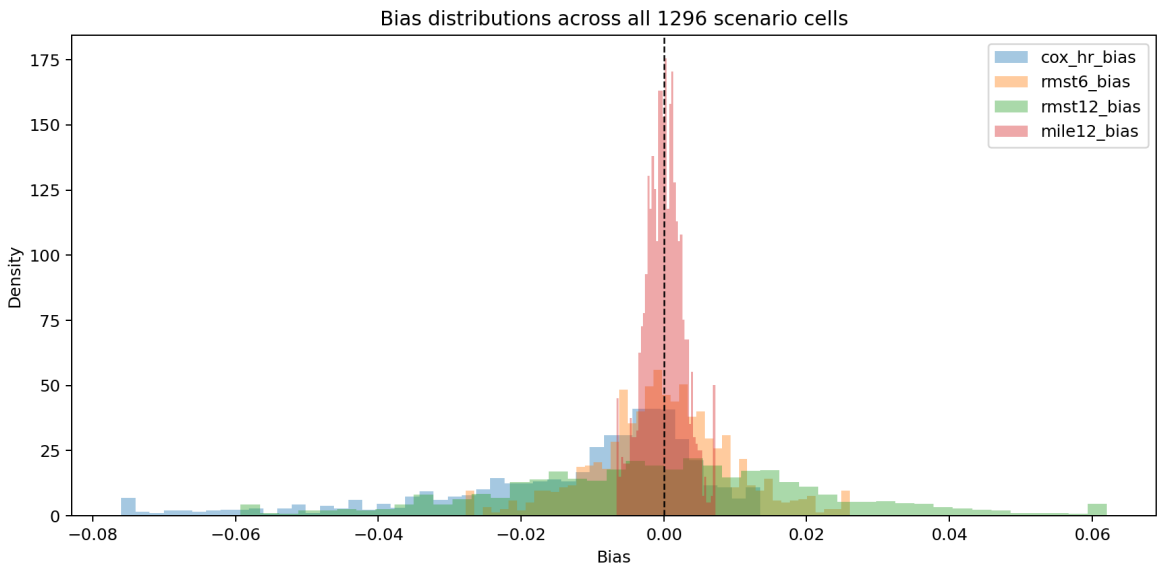

Figure 3: Bias distributions across all scenario cells.

3.2 One-factor-at-a-time slices for interpretability

Reference scenario used for OFAT plots: $n = 500$, $\lambda_0 = 0.057762$, censoring= 0.1, prevalence= 0.3, $hr_{overall} = 0.8$, relative effect= 0.8.

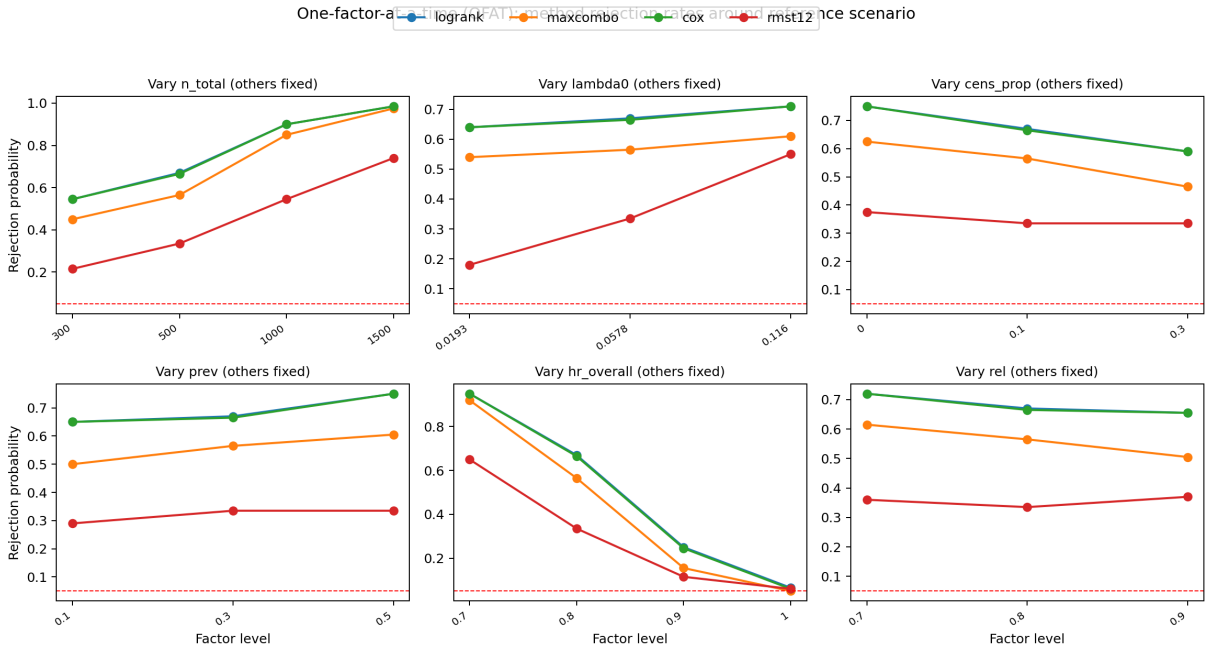

Figure 4: OFAT rejection-rate profiles: each panel varies one factor while others remain at the reference scenario.

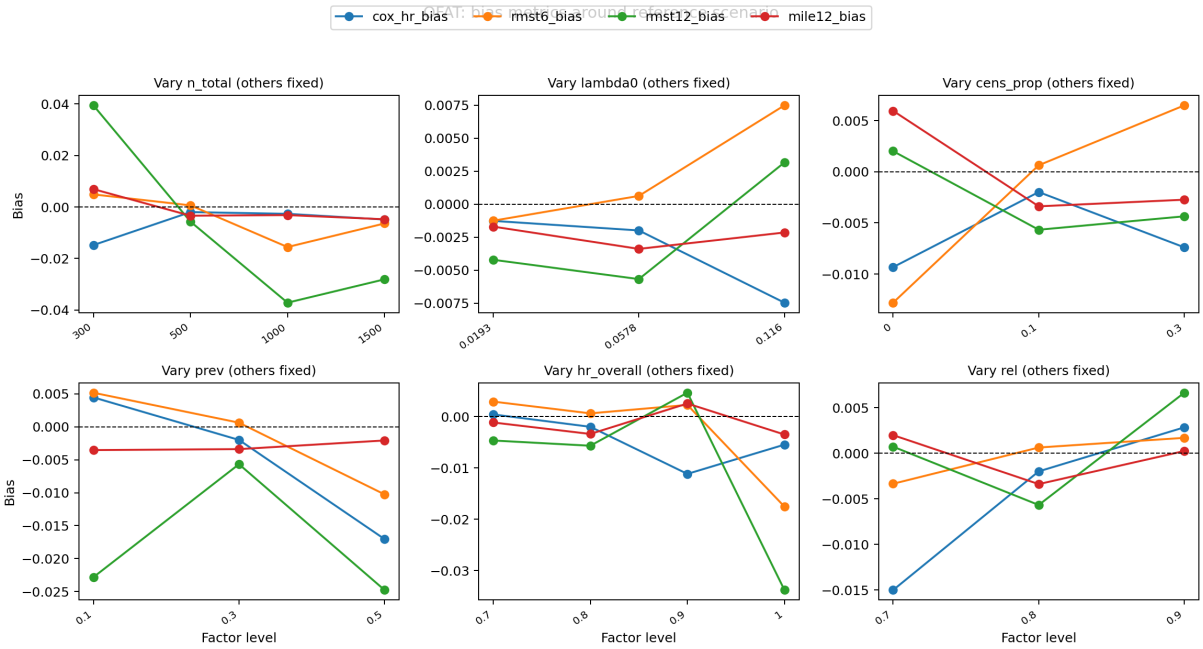

Figure 5: OFAT bias profiles for key estimators.

Sample size	Log-rank rej	MaxCombo rej	Cox rej	RMST12 rej	Cox bias	RMST12 bias
300	0.385	0.314	0.383	0.204	-0.0096	-0.0002
500	0.493	0.430	0.492	0.282	-0.0125	0.0006
1000	0.610	0.565	0.610	0.406	-0.0140	-0.0010
1500	0.669	0.630	0.669	0.478	-0.0151	0.0005

Table 1: Summary by per-replication sample size (all other factors averaged).

4 Discussion

The full-grid analysis confirms strong scenario dependence: no single method dominates uniformly. RMST methods remain interpretable under marginal NPH, while weighted log-rank variants and MaxCombo provide improved detection in many non-proportional settings. OFAT slices greatly improve recoverability and practical interpretation compared with fully aggregated displays alone.

5 Conclusions

The study now combines (i) full-grid completeness, (ii) explicit scenario recoverability via scenario IDs and supplementary CSVs, and (iii) OFAT interpretation plots. This combination supports both transparency and decision-oriented reading.

References

[1] D. R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2):187–220, 1972.

[2] T. R. Fleming and D. P. Harrington. *Counting Processes and Survival Analysis*. Wiley, 1991.

[3] D. P. Harrington and T. R. Fleming. A class of rank test procedures for censored survival data. *Biometrika*, 69(3):553–566, 1982.

[4] D. Magirr and C. F. Burman. Modestly weighted logrank tests. *Statistics in Medicine*, 38(20): 3782–3790, 2019.

[5] T. P. Morris, I. R. White, and M. J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019.

[6] R. Ristl et al. nph: An r package for non-proportional hazards methods and simulation. *R package documentation*, 2021.

[7] P. Royston and M. K. B. Parmar. Flexible parametric proportional-hazards and proportional-odds models for censored survival data. *Statistics in Medicine*, 21(15):2175–2197, 2002.

[8] M. Schemper, S. Wakounig, and G. Heinze. The estimation of average hazard ratios by weighted cox regression. *Statistics in Medicine*, 28(19):2473–2489, 2009.

[9] H. Uno, B. Claggett, L. Tian, et al. Moving beyond the hazard ratio in quantifying the between-group difference in survival analysis. *Journal of Clinical Oncology*, 32(22):2380–2385, 2014.

[10] H. Uno, L. Tian, et al. survrm2: Comparing restricted mean survival time. *R package vignette*, 2021.